

PROJECT PRESENTATION & REPORT

(CMPE343)

MEMBER S

ZAKARIA EL MILKI (22321066): Observed library systems, created the ER-diagram and managed the DDL and DML files in order to visualize a library's system.

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INTRODUCTION

The following project implements a relational database for Ercan Airport, managing airplanes, technical staff, traffic controllers, hangar storage, and safety inspections. It ensures data integrity through complex relationships and foreign keys, supporting queries that help track aircraft maintenance, monitor technician expertise, analyze hangar occupancy, and manage airport operations efficiently.

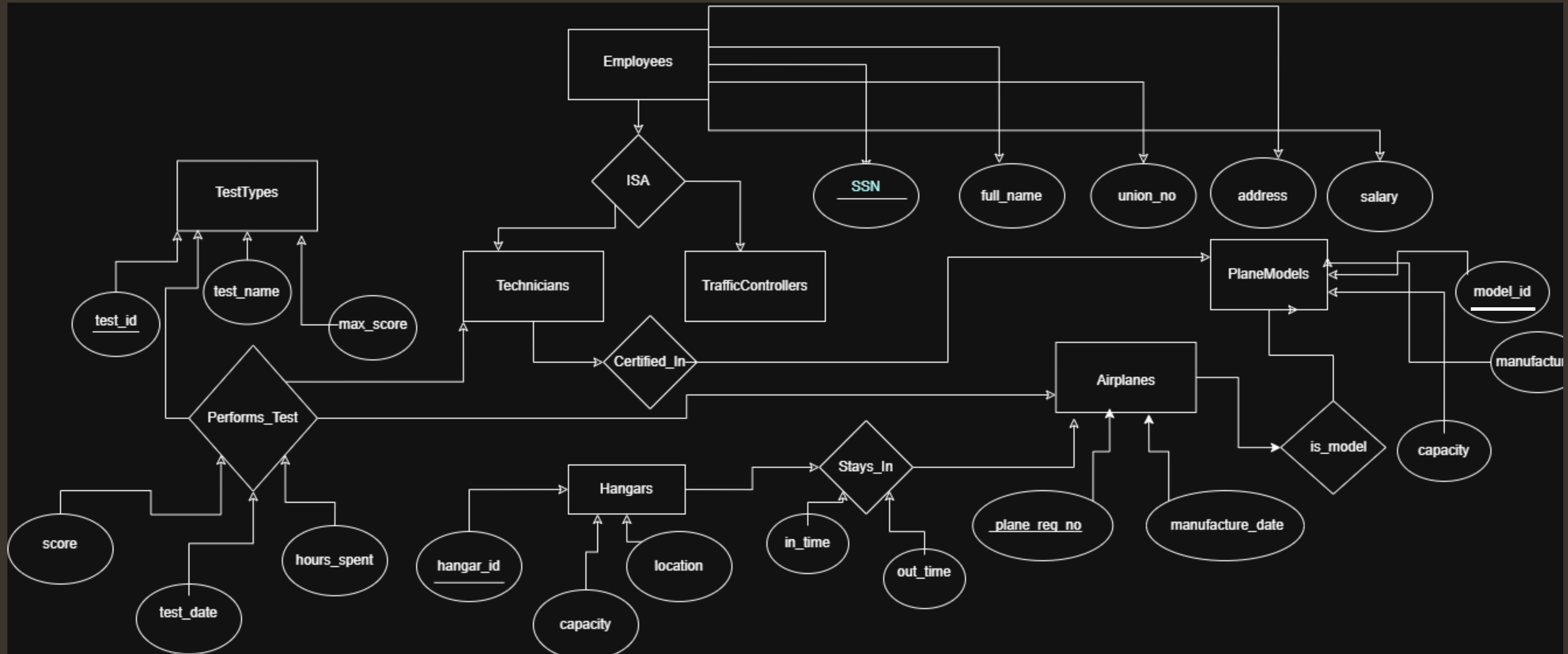
OVERALL STRUCTURE

- **Employees** (ssn (PK), full_name, address, union_membership_no, salary)
- **Technicians** (ssn (PK/FK), rank)
- **TrafficControllers** (ssn (PK/FK), last_exam_date)
- **PlaneModels** (model_id (PK), manufacturer, capacity)
- **Airplanes** (plane_reg_no (PK), model_id (FK))
- **Hangars** (hangar_id (PK), location, capacity)
- **HangarLogs** (log_id (PK), plane_reg_no (FK), hangar_id (FK), in_time, out_time)
- **TestTypes** (test_id (PK), test_name, max_score)
- **Expertise** (ssn (PK/FK), model_id (PK/FK))
- **TestRecords** (event_id (PK), ssn (FK), plane_reg_no (FK), test_id (FK), test_date, hours_spent, score)

RELATIONSHIPS BETWEEN ENTITIES

- Employees and Roles: Technicians and Traffic Controllers are specialized types of Employees (1:1).
- Models and Airplanes: Each specific airplane belongs to one model, but a model can have many airplanes (1:N).
- Expertise: Technicians are certified for specific plane models (N:N).
- Hangar Tracking: An airplane is logged into a hangar for a specific duration (1:N).
- Testing: Technicians use specific tests to check airplanes, creating a detailed record of the event.

ENTITY RELATIONSHIP DIAGRAM (ERD)



DDL (MANAGING DATABASE AND TABLES)

```
1  -- 1. STORES GENERAL INFORMATION FOR ALL STAFF (BASIC INFO)
2  CREATE TABLE Employees (
3      ssn INT PRIMARY KEY,
4      full_name VARCHAR(100) NOT NULL,
5      address VARCHAR(255),
6      union_membership_no INT UNIQUE,
7      salary DECIMAL(10, 2)
8  );
9
10 -- 2. STORES THE SPECIFIC RANK FOR MAINTENANCE STAFF
11 CREATE TABLE Technicians (
12     ssn INT PRIMARY KEY,
13     rank VARCHAR(50),
14     FOREIGN KEY (ssn) REFERENCES Employees(ssn) ON DELETE CASCADE
15 );
16
17 -- 3. STORES MEDICAL EXAM DATES FOR THE CONTROL TOWER STAFF
18 CREATE TABLE TrafficControllers (
19     ssn INT PRIMARY KEY,
20     last_exam_date DATE,
21     FOREIGN KEY (ssn) REFERENCES Employees(ssn) ON DELETE CASCADE
22 );
23
24 -- 4. STORES TECHNICAL SPECS (BLUEPRINTS) FOR PLANE TYPES
25 CREATE TABLE PlaneModels (
26     model_id INT PRIMARY KEY,
27     manufacturer VARCHAR(100),
28     capacity INT
29 );
30
31 -- 5. STORES THE PHYSICAL PLANES CURRENTLY AT THE AIRPORT
32 CREATE TABLE Airplanes (
33     plane_reg_no VARCHAR(20) PRIMARY KEY,
34     model_id INT,
35     manufacture_date DATE,
36     FOREIGN KEY (model_id) REFERENCES PlaneModels(model_id)
37 );
```

```
-- 6. STORES PHYSICAL STORAGE LOCATIONS (PARKING BAYS)
CREATE TABLE Hangars (
    hangar_id INT PRIMARY KEY,
    location VARCHAR(100),
    capacity INT
);

-- 7. TRACKS THE HISTORY OF PLANES MOVING IN AND OUT OF HANGARS
CREATE TABLE HangarLogs (
    log_id INT PRIMARY KEY AUTO_INCREMENT,
    plane_reg_no VARCHAR(20),
    hangar_id INT,
    in_time DATETIME,
    out_time DATETIME,
    FOREIGN KEY (plane_reg_no) REFERENCES Airplanes(plane_reg_no),
    FOREIGN KEY (hangar_id) REFERENCES Hangars(hangar_id)
);

-- 8. DEFINES THE DIFFERENT SAFETY TESTS AVAILABLE (NAME AND MAX SCORE)
CREATE TABLE TestTypes (
    test_id INT PRIMARY KEY,
    test_name VARCHAR(100),
    max_score INT
);

-- 9. LINKS TECHNICIANS TO THE PLANE MODELS THEY ARE CERTIFIED TO FIX
CREATE TABLE Expertise (
    ssn INT,
    model_id INT,
    PRIMARY KEY (ssn, model_id),
    FOREIGN KEY (ssn) REFERENCES Technicians(ssn),
    FOREIGN KEY (model_id) REFERENCES PlaneModels(model_id)
);
```

```
73  -- 10. THE MASTER LOG: TRACKS WHICH TECH TESTED WHICH PLANE AND THE SCORE
74  CREATE TABLE TestRecords (
75      event_id INT PRIMARY KEY AUTO_INCREMENT,
76      ssn INT,
77      plane_reg_no VARCHAR(20),
78      test_id INT,
79      test_date DATE,
80      hours_spent INT,
81      score INT,
82      FOREIGN KEY (ssn) REFERENCES Technicians(ssn),
83      FOREIGN KEY (plane_reg_no) REFERENCES Airplanes(plane_reg_no),
84      FOREIGN KEY (test_id) REFERENCES TestTypes(test_id)
85  );
```


DML STATEMENTS

```
1  -- 1. INSERTING EMPLOYEES (General Info)
2  INSERT INTO Employees (ssn, full_name, address, union_membership_no, salary) VALUES
3  (101, 'Ahmet Yilmaz', 'Lefkosa, Cyprus', 5001, 45000.00),
4  (102, 'Ayse Demir', 'Girne, Cyprus', 5002, 48000.00),
5  (103, 'John Smith', 'Magusa, Cyprus', 5003, 42000.00),
6  (104, 'Fatma Avci', 'Lefkosa, Cyprus', 5004, 55000.00),
7  (105, 'Mehmet Kaya', 'Guzelyurt, Cyprus', 5005, 43000.00);
8
9  -- 2. INSERTING TECHNICIANS (Linking to Employees)
10 INSERT INTO Technicians (ssn, rank) VALUES
11 (101, 'Senior Engineer'),
12 (103, 'Junior Mechanic'),
13 (105, 'Lead Technician');
14
15 -- 3. INSERTING TRAFFIC CONTROLLERS (Linking to Employees)
16 INSERT INTO TrafficControllers (ssn, last_exam_date) VALUES
17 (102, '2026-01-15'),
18 (104, '2025-12-10');
19
20 -- 4. INSERTING PLANE MODELS
21 INSERT INTO PlaneModels (model_id, manufacturer, capacity) VALUES
22 (737, 'Boeing', 180),
23 (320, 'Airbus', 150),
24 (777, 'Boeing', 350);
25
26 -- 5. INSERTING ACTUAL AIRPLANES
27 INSERT INTO Airplanes (plane_reg_no, model_id, manufacture_date) VALUES
28 ('TC-ERC01', 737, '2018-05-20'),
29 ('TC-ERC02', 320, '2020-11-12'),
30 ('TC-ERC03', 777, '2015-03-25');
31
32 -- 6. INSERTING HANGARS
33 INSERT INTO Hangars (hangar_id, location, capacity) VALUES
34 (1, 'North Wing - Bay A', 5),
35 (2, 'South Wing - Bay B', 3);
36
```

```
32 -- 6. INSERTING HANGARS
33 INSERT INTO Hangars (hangar_id, location, capacity) VALUES
34 (1, 'North Wing - Bay A', 5),
35 (2, 'South Wing - Bay B', 3);
36
37 -- 7. INSERTING EXPERTISE (Which Tech knows which Model)
38 INSERT INTO Expertise (ssn, model_id) VALUES
39 (101, 737),
40 (101, 777),
41 (105, 320),
42 (105, 737);
43
44 -- 8. INSERTING TEST TYPES
45 INSERT INTO TestTypes (test_id, test_name, max_score) VALUES
46 (1, 'Engine Pressure Test', 100),
47 (2, 'Hydraulic Fluid Check', 100),
48 (3, 'Avionics Calibration', 50);
49
50 -- 9. INSERTING TEST RECORDS (The actual events)
51 INSERT INTO TestRecords (ssn, plane_reg_no, test_id, test_date, hours_spent, score) VALUES
52 (101, 'TC-ERC01', 1, '2026-04-10', 5, 95),
53 (105, 'TC-ERC02', 2, '2026-04-12', 3, 88),
54 (101, 'TC-ERC03', 3, '2026-04-15', 8, 48);
55
56 -- 10. INSERTING HANGAR LOGS (Parking History)
57 INSERT INTO HangarLogs (plane_reg_no, hangar_id, in_time, out_time) VALUES
58 ('TC-ERC01', 1, '2026-04-01 08:00:00', '2026-04-05 12:00:00'),
59 ('TC-ERC02', 2, '2026-04-10 09:30:00', NULL); -- NULL means it's still inside
```

QUERIE S

The following queries that will be displayed alongside with their output benefit much in filtering data in library database which outlines the important role of such techniques in facilitating the work and process of complex systems

-- Q1: LIST ALL EMPLOYEES AND THEIR SALARIES

SELECT full_name, salary FROM
Employees;

full_name	salary
Ahmet Yilmaz	45000.00
Ayse Demir	48000.00
John Smith	42000.00
Fatma Avcı	55000.00
Mehmet Kaya	43000.00

-- Q2: FIND ALL TECHNICIANS (JOINING EMPLOYEES AND TECHNICIANS)

```
SELECT E.full_name, T.rank
FROM Employees E
JOIN Technicians T ON E.ssn =
      T.ssn;
```

full_name	rank
Ahmet Yilmaz	Senior Engineer
John Smith	Junior Mechanic
Mehmet Kaya	Lead Technician

-- Q3: FIND PLANES CURRENTLY PARKED IN HANGAR 1 (WHERE
OUT_TIME IS NULL)

```
SELECT plane_reg_no, in_time  
FROM HangarLogs  
WHERE hangar_id = 1 AND out_time IS  
NULL;
```

plane_reg_no	in_time
TC-ERC02	2026-04-10 09:30:00

-- Q4: LIST PLANE MODELS WITH CAPACITY OVER 200

SELECT manufacturer, capacity
FROM PlaneModels
WHERE capacity > 200;

SEATS	
manufacturer	capacity
Boeing	350

```
-- Q5: FIND ALL EXPERTS CERTIFIED ON THE 'BOEING  
      737'  
SELECT E.full_name  
FROM Employees E  
JOIN Expertise EX ON E.ssn =  
      EX.ssn  
WHERE EX.model_id = 737;
```

full_name
Ahmet Yilmaz

-- Q6: SHOW MAINTENANCE HISTORY
FOR A SPECIFIC PLANE (TC-ERC01)

```
SELECT T.test_name, TR.test_date, TR.score  
FROM TestRecords TR  
JOIN TestTypes T ON TR.test_id = T.test_id  
WHERE TR.plane_reg_no = 'TC-ERC01';
```

test_name	test_date	score
Engine Test	2026-04-10	95

-- Q7: COUNT HOW MANY PLANES EACH HANGAR IS CURRENTLY
HOLDING

```
SELECT hangar_id, COUNT(*) AS  
       planes_inside  
FROM HangarLogs  
WHERE out_time IS NULL  
GROUP BY hangar_id;
```

hangar_id	planes_inside
2	1

-- Q8: FIND THE AVERAGE SALARY OF ALL AIRPORT
EMPLOYEES

```
SELECT AVG(salary) AS  
average_airport_salary  
FROM Employees;
```

average_airport_salary
46600.000000

-- Q9: LIST PLANES THAT HAVE NEVER HAD A SAFETY TEST (CRITICAL FOR SAFETY)

```
SELECT plane_reg_no
FROM Airplanes
WHERE plane_reg_no NOT IN
(SELECT plane_reg_no FROM
TestRecords);
```

plane_reg_no
TC-ERC03

-- Q10: FIND TRAFFIC CONTROLLERS WHO HAVEN'T HAD AN EXAM SINCE
2025

```
SELECT E.full_name,  
       TC.last_exam_date  
FROM Employees E  
JOIN TrafficControllers TC ON E.ssn =  
       TC.ssn  
WHERE TC.last_exam_date < '2026-01-  
       01';
```

+	-----	+	-----	+
	full_name		last_exam_date	
+	-----	+	-----	+
	Fatma Avcı		2025-12-10	
+	-----	+	-----	+

-- Q11: CALCULATE TOTAL HOURS SPENT ON TESTS BY EACH

TECHNICIAN
SELECT E.full_name, SUM(TR.hours_spent) AS
total_hours
FROM Employees E
JOIN TestRecords TR ON E.ssn = TR.ssn
GROUP BY E.full_name;

full_name	total_hours
Ahmet Yilmaz	5
Mehmet Kaya	3

```
-- Q12: SHOW THE HIGHEST SCORE ACHIEVED ON EACH TYPE OF
SELECT TT.test_name, MAX(TR.score)    TEST
      AS top_score
FROM TestTypes TT
JOIN TestRecords TR ON TT.test_id =
      TR.test_id
GROUP BY TT.test_name;
```

test_name	top_score
Engine Test	95
Hydraulic Check	88

-- Q13: FIND TECHNICIANS WHO ARE EXPERTS IN MORE THAN 1
MODEL

```
SELECT E.full_name,  
COUNT(EX.model_id) AS models_known  
FROM Employees E  
JOIN Expertise EX ON E.ssn = EX.ssn  
GROUP BY E.full_name  
HAVING COUNT(EX.model_id) > 1;
```

full_name	models_known
Ahmet Yilmaz	2

-- Q14: JOIN EMPLOYEES, TESTRECORDS, AND TESTTYPES (3-WAY JOIN)

```
SELECT E.full_name AS Technician,  
       TT.test_name, TR.score  
FROM Employees E  
JOIN TestRecords TR ON E.ssn =  
       TR.ssn  
JOIN TestTypes TT ON TR.test_id =  
       TT.test_id;
```

Technician	test_name	score
Ahmet Yilmaz	Engine Test	95
Mehmet Kaya	Hydraulic Check	88

-- Q15: COMPARE TOTAL SALARY BUDGET: TECHNICIANS VS CONTROLLERS

```
SELECT 'Technician' AS Role, SUM(salary) FROM Employees
WHERE ssn IN (SELECT ssn FROM Technicians)
UNION
```

```
SELECT 'Controller', SUM(salary) FROM Employees WHERE ssn IN
(SELECT ssn FROM TrafficControllers);
```

Role	SUM(salary)
Technician	130000.00
Controller	103000.00

CONCLUSIO

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The design and implementation of the Ercan Airport database successfully demonstrate how a relational model can streamline complex aviation operations.

By organizing data into ten normalized tables, we have created a system that ensures data integrity while providing deep insights into employee management, technician certifications, and aircraft maintenance history. The 15 SQL queries developed for this project serve as powerful management tools, allowing for real-time tracking of hangar capacity and safety test performance. Overall, this database provides a scalable and robust foundation that can be easily expanded to include flight scheduling and passenger services in the future.

THANK
YOU

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